

Project Design Phase

Problem - Solution Fit

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

Purpose: confirm that AI StudyBuddy solves a real problem for its target students and understand the current situation in order to improve it.

Canvas block	AI StudyBuddy
1. Customer segment(s)	College and school students preparing for semester, competitive and placement examinations; administrators of the platform
2. Jobs to be done / problems	Summarize long material; make flashcards and quizzes; build a study plan; keep all resources in one place
3. Triggers to act	Approaching examinations, large volume of content, limited preparation time, need to check understanding
4. Emotions (before / after)	Before: overwhelmed, stressed, unsure. After: organized, confident, in control of revision
5. Available solutions	Manual note-taking, separate flashcard and quiz apps, generic chat assistants with no saved study library
6. Customer constraints	Limited time, scattered file formats, need for secure accounts, dependence on a reliable AI service
7. Root cause of the problem	Note making, self-testing and scheduling are manual, repetitive and spread across several tools
8. Behaviour	Students copy content between apps, write notes by hand and create schedules in spreadsheets or on paper
9. Channels of behaviour	Online: web app (React frontend) calling REST APIs, tested with Postman. Offline: handwritten notes, printed textbooks
10. Solution	A secure Node.js/Express backend with JWT and role-based access that stores uploaded material in MongoDB and uses Google Gemini to generate summaries, flashcards, quizzes and study plans